CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Fourth Semester of B. Pharm. Examination
University Theory Examination April 2015
PH211 Pharmaceutical Analysis - II

Date: 24.04.2015, Friday **Time:10.00 a.m. to 01.00 p.m.** **Maximum Marks: 80**

Instructions:
1. **Figures to right indicate marks.**
2. **Section wise separate answer book to be used.**
3. **Draw neat sketches wherever necessary.**

SECTION- I

Q 1		Attempt any **TWO** of the following	
A	**(a)**	Explain the term Analysis. Classify the modern techniques of instrumental analysis and give atleast two examples of technique in each class.	**05**
	(b)	Write advantages and limitations of Instrumental methods of analysis.	**05**
B	**(a)**	Define the following term. 1) Chromatography 2) Peak Resolution 3) Rf 4) HETP 5) Partitioning	**05**
	(b)	Define DSC. Explain the principle of DSC technique.	**05**
C		Write a detail note on Thin Layer Chromatography.	**10**
Q 2		Attempt any **TWO** of the following	
A	**(a)**	Explain different types of paper chromatography.	**05**
	(b)	Enlist the adsorbents used as stationary phase in column chromatography. Write about applications of column chromatography.	**05**
B		Discuss principle, instrumentation and applications of DTA.	**10**
C	**(a)**	Discuss factors affecting TGA analysis.	**05**
	(b)	How will you calibrate pH meter and conductometer?	**05**

SECTION- II

Q 3		Attempt any **TWO** of the following	
A	**(a)**	Define followings: 1) Electrode 2) Specific Optical Rotation 3) Ohm's law 4) Reference electrode 5) Potential	**05**

	(b)	Describe different methods of detecting end points in potentiometric titrations.	**05**
B		What is conductometric titration? Explain its principle. Give example of different types of conductometric titrations and draw shape of titration curve for each example.	**10**
C		Enlist reference and indicator electodes used in Potentiometry. Describe ANY TWO reference electrodes with respect to design, advantages and disadvantages.	**10**
Q 4		Attempt any **TWO** of the following	
A	**(a)**	Write applications of Polarography	**05**
	(b)	Write short note on Amperometry.	**05**
B		Write about design and features of following electrodes. 1) Glass electrode 2) Dropping mercury electrode	**10**
C		Write a note on principle, instrumentation and applications of Polarimetry.	**10**